

ARGOS: initial investment estimate

Working budget for discussion

13 September 2026

1 Scope and proposed funding

The proposed programme builds one PET scanner, one Compton camera and the integrated software and service needed for a full system demonstrator. Development with Donostia Hospital proceeds from phantom tests to clinical investigations, clinical validation and conformity assessment.

The central estimate is **€ 6.7 million over four years**, with a **funding target of € 7 million**. A first investment of € 2–3 million could finance the initial stage, but would not cover the complete programme and company operations.

The estimate assumes a Donostia-based team, access to hospital and proton-beam facilities by agreement, and no construction of a proton-therapy facility. Costs are planning allowances in 2026 euros, excluding recoverable VAT. No sales revenue, grants or partner contributions are deducted. Price increases and schedule overruns draw on the programme reserve.

2 Four-year budget

The raw-hardware estimate comprises € 740k for PET materials, € 100k for the Compton camera and € 160k for computing and other hardware. These inputs follow the existing installation model.

The € 200k prototype allowance covers external purchases and contracted work. Employee assembly, integration and software development are charged to personnel. The full delivered-system cost is not added again. The first-device contingency is 25% of the € 1.2M hardware and external-work base; the programme reserve applies only to the remaining expenditure.

Working capital is a cash requirement, not an additional development expense. The difference between € 6.66M and the € 7M funding target provides € 340k of further headroom.

3 Personnel

Annual figures are total employer costs, including employer contributions, not gross salaries. They are proposed hiring allowances rather than salary survey results. Founders taking operating roles are included in the paid team.

The software engineers develop reconstruction, data integration, interfaces and service tools. Their costs are company development expenditure; they are not included in the non-personnel IT allowance.

The medical physicist leads validation. A separate monitoring specialist operates the daily clinical service during the two clinical years. The € 80k/year provision follows the existing monitoring-cost assumption, including relief coverage for one shift. Hospital staff and trial costs are in the separate

Item	€ k	Scope
PET, Compton camera and computing	1,000	Existing raw-hardware estimate.
Prototype fabrication and test equipment	200	Purchased fabrication, phantoms, fixtures, installation and external technical work.
First-device contingency	300	Failed components, redesigns and replacement parts.
Personnel	2,660	Development, management, quality and clinical monitoring.
Hospital and clinical programme	500	Phantom sessions, beam access, hospital staff, trial coordination, data management and study insurance.
External regulatory and conformity work	250	Specialist advice, independent testing, audits and certification work.
Office and laboratory space	240	€ 60k/year, including utilities.
Software tools and IT operations	120	€ 30k/year: tools, security, backup and licences.
Company administration	160	€ 40k/year: accounting, legal work, IP and corporate insurance.
Travel and transport	120	€ 30k/year: hospital visits, suppliers, conferences and equipment transport.
Planned expenditure	5,550	Includes the first-device contingency.
Programme reserve	810	20% of the € 4.05M outside the hardware/prototype package.
Working-capital buffer	300	Payment timing, recoverable VAT and cash at programme completion.
Funding requirement	6,660	Round to € 6.7M; target € 7M.

Role	Staffing period	Annual € k	Total € k
CEO / project and business lead	1 × 4 years	90	360
System architect / technical lead	1 × 4 years	100	400
Detector and electronics engineer	1 × 4 years	85	340
Software engineers	2 × 4 years	85 each	680
Medical physicist / validation lead	1 × 4 years	90	360
Assembly and laboratory technician	1 × 3 years	60	180
Quality and regulatory lead	0.5 × 4 years	90 per FTE	180
Clinical monitoring specialist	1 × 2 years	80	160
Total			2,660

€ 500k clinical allowance, which should be replaced by a costed hospital agreement once the protocol and patient numbers are defined.

4 Development sequence

Year 1. Detailed design, procurement, software architecture, quality system and clinical protocol.

Year 2. Integrated prototype, phantom measurements, reliability testing and service procedures.

Years 3–4. Authorised clinical investigations, system improvements, clinical validation and conformity assessment.

Donostia Hospital is the clinical development and validation partner. Device market certification follows the applicable conformity-assessment route; the hospital does not itself grant it. Spanish clinical investigations require the relevant ethics review and AEMPS authorisation [1]; notified bodies assess conformity where required by the device’s regulatory route [2].

The funded endpoint is a working, clinically evaluated demonstrator and its service, with European conformity work included. The budget does not cover a large multicentre study proving improved patient outcomes, worldwide market approvals or production facilities for commercial-scale manufacturing.

5 Effect of programme duration

Development period	Funding range
3 years	€ 5.5–6M
4 years	€ 6.5–7M
5 years	€ 7.5–8M

These are indicative duration variants, not separately costed programmes. The three-year case requires substantial design maturity and prompt access to clinical testing; compressing the schedule does not automatically reduce engineering effort. Retaining the team and facilities for an additional year costs approximately € 0.8–0.9M before contingency.

The proposed starting point is **€ 7M over four years**, potentially raised in stages against prototype and clinical milestones. This note records the budget proposal separately from the six existing financial and market tables; it does not alter the presentation or workbook.

References

- [1] AEMPS, *Investigaciones clínicas con productos sanitarios*. <https://www.aemps.gob.es/productos-sanitarios/investigacionclinica-productossanitarios/>.
- [2] European Commission, *Notified bodies for medical devices*. https://health.ec.europa.eu/medical-devices-topics-interest/notified-bodies-medical-devices_en.